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Formatting guide

This guide describes how to prepare contributions for submission. We recommend you read this in full if you have not previously submitted a contribution to *Nature*. We also recommend that, before submission, you familiarize yourself with *Nature*'s style and content by reading the journal, either in print or online, particularly if you have not submitted to the journal recently.

Formats for *Nature* contributions

Articles are the main format for original research contributions to *Nature*. In addition, *Nature* publishes other submitted material as detailed below.

Articles

Articles are original reports whose conclusions represent a substantial advance in understanding of an important problem and have immediate, far-reaching implications. In print, physical sciences papers do not normally exceed **6 pages** on average, and biological, clinical and social-sciences papers do not normally exceed **8 pages** on average. However, the final print length is at the editor's discretion.

Articles start with a fully referenced summary paragraph, ideally of no more than 200 words, which is separate from the main text and avoids numbers, abbreviations, acronyms or measurements unless essential. It is aimed at readers outside the discipline. This summary paragraph should be structured as follows: 2-3 sentences of basic-level introduction to the field; a brief account of the background and rationale of the work; a statement of the main conclusions (introduced by the phrase 'Here we

show' or its equivalent); and finally, 2-3 sentences putting the main findings into general context so it is clear how the results described in the paper have moved the field forwards. Please refer to our annotated example to see how the summary paragraph should be constructed.

The typical length of a 6-page article with 4 modest display items (figures and tables) is 2500 words (summary paragraph plus body text). The typical length of an 8-page article with 5-6 modest display items is 4300 words. A 'modest' display item is one that, with its legend, occupies about a quarter of a page (equivalent to ~270 words). If a composite figure (with several panels) needs to occupy at least half a page in order for all the elements to be visible, the text length may need to be reduced accordingly to accommodate such figures. Keep in mind that essential but technical details can be moved into the Methods or Supplementary Information.

As a guideline, articles typically have no more than 50 references. (There is no such constraint on any additional references associated with Methods or Supplementary Information.)

Sections are separated with subheadings to aid navigation. Subheadings may be up to 40 characters (including spaces).

Word counts refer to the text of the paper. Title, author list, acknowledgements and references are not included in total word counts.

Matters Arising and Corrections

Matters Arising are exceptionally interesting or important comments and clarifications on original research papers or other peer-reviewed material published within the past 18 months in *Nature*. They are published online but not in print.

For further details of and instructions for how to submit such comments on peer-reviewed material published in *Nature* — or to notify editors of

the potential need for a correction — please consult our Matters Arising page.

Other contributions to *Nature*

Please access the other submitted material pages for further details on any of the contribution types below:

- News and Comment
- Correspondence
- Obituaries
- Opinion
- Books & Arts
- Futures
- News & Views
- Insights, Reviews and Perspectives
- Analysis
- Hypothesis
- Careers
- Technology Features
- Outlooks

The editorial process

See this section for an explanation of *Nature*'s editorial criteria for

publication, refereeing policy and how editors handle papers after submission. Submission to a *Nature* journal is taken by the journal to mean that all the listed authors have agreed to all of the contents. See authorship policy for more details.

Presubmission enquiries

If you wish to enquire whether your Article might be suitable for consideration by *Nature*, please use our online presubmission enquiry service. All presubmission enquiries must include a cover paragraph to the editor stating the interest to a broad scientific readership, a fully referenced summary paragraph, and a reference list.

Readability

Nature is an international journal covering all the sciences. Contributions should therefore be written clearly and simply so that they are accessible to readers in other disciplines and to readers for whom English is not their first language. Thus, technical jargon should be avoided as far as possible and clearly explained where its use is unavoidable. Abbreviations, particularly those that are not standard, should also be kept to a minimum. The background, rationale and main conclusions of the study should be clearly explained. Titles and abstracts in particular should be written in language that will be readily intelligible to any scientist. Essential but specialized terms should be explained concisely but not didactically.

For gene, protein and other specialized names authors can use their preferred terminology so long as it is in current use by the community, but they must give all known names for the entity at first use in the paper. *Nature* prefers authors to use internationally agreed nomenclature. Papers containing new or revised formal taxonomic nomenclature for animals, whether living or extinct, are accepted conditional on the provision of LSIDs (Life Science Identifiers) by means of registration of such nomenclature with ZooBank, the proposed online registration

system for the International Code of Zoological Nomenclature (ICZN).

Even though no paper will be rejected because of poor language, non–native English speakers occasionally receive feedback from editors and reviewers regarding language and grammar usage in their manuscripts. You may wish to consider asking colleagues to read your manuscript and/or to use a professional editing service such as those provided by our affiliates *Nature* Research Editing Service or American Journal Experts. You can also get a fast, free grammar check of your manuscript that takes into account all aspects of readability in English. Please note that the use of a language editing service is not a requirement for publication in *Nature*.

Nature's editors provide detailed advice about the expected print length when asking for the final version of the manuscript. *Nature*'s editors often suggest revised titles and rewrite the summary paragraphs of Articles so the conclusions are clear to a broad readership.

After acceptance, *Nature*'s subeditors (copyeditors) ensure that the text and figures are readable and clear to those outside the field, and edit papers into *Nature*'s house style. They pay particular attention to summary paragraphs, overall clarity, figures, figure legends and titles.

Proofs are sent before publication; authors are welcome to discuss proposed changes with *Nature*'s subeditors, but *Nature* reserves the right to make the final decision about matters of style and the size of figures.

A useful set of articles providing general advice about writing and submitting scientific papers can be found on the SciDev.Net website.

Format of Articles

Contributions should be double-spaced and written in English (spellings as in the Oxford English Dictionary).

Contributions should be organized in the sequence: title, authors, affiliations (plus present addresses), bold first paragraph, main text, main references, tables, figure legends, methods (including separate data and code availability statements), methods references, acknowledgements, funding statement, author contributions, competing interest declaration, additional information (containing supplementary information line (if any) and corresponding author line), extended data figure/table legends. In order to facilitate the review process, for initial submissions we encourage authors to present the manuscript text and figures together in a single file (Microsoft Word or PDF, up to 30 MB in size). The figures may be inserted within the text at the appropriate positions or grouped at the end, and each figure legend should be presented together with its figure. Also, please include line numbers within the text.

Titles

Titles do not exceed two lines in print. This equates to 75 characters (including spaces). Titles do not normally include numbers, acronyms, abbreviations or punctuation. They should include sufficient detail for indexing purposes but be general enough for readers outside the field to appreciate what the paper is about.

Text

An uninterrupted page of text contains about 1250 words.

- A typical 6-page Article contains about 2,500 words of text and, additionally, 4 modest display items (figures and/or tables) with brief legends, reference list and online-only methods section if applicable. A composite figure (with several panels) usually needs to take about half a page, equivalent to about 600 words, in order for all the elements to be visible (see section 5.9 for instructions on sizing figures).

- A typical 8-page Article contains about 4300 words of text and, additionally, 5-6 modest display items (figures and/or tables) with

brief legends, reference list and online-only methods section if applicable. A composite figure (with several panels) usually needs to take about half a page, equivalent to about 600 words, in order for all the elements to be visible (see section 5.9 for instructions on sizing figures).

Authors of contributions that significantly exceed the limits stated here (or as specified by the editor) will have to shorten their papers before acceptance, inevitably delaying publication.

Nature requires authors to specify the contribution made by their co-authors in the end notes of the paper (see section 5.5). If authors regard it as essential to indicate that two or more co-authors are equal in status, they may be identified by an asterisk symbol with the caption 'These authors contributed equally to this work' immediately under the address list. If more than three co-authors are equal in status, this should be indicated in the author contributions statement. Present addresses appear immediately below the author list (below the footnote rule at the bottom of the first page) and may be identified by a dagger symbol; all other essential author-related explanation is placed in the acknowledgements.

Our preferred format for text is Microsoft Word, with the style tags removed.

TeX/LaTeX: If you have prepared your paper using TeX/LaTeX, we will need to convert this to Word after acceptance, before your paper can be typeset. All textual material of the paper (including references, tables, figure captions, online methods, etc.) should be included as a single .tex file.

We prefer the use of a 'standard' font, preferably 12-point Times New Roman. For mathematical symbols, Greek letters and other special characters, use normal text or Symbol font. Word Equation Editor/

MathType should be used only for formulae that cannot be produced using normal text or Symbol font.

Methods

The 'Methods' section is in the main text file, following the figure legends. This Methods section will appear in the PDF and in the full-text (HTML) version of the paper online, but will not appear in the printed issue. The Methods section should be written as concisely as possible but should contain all elements necessary to allow interpretation and replication of the results. As a guideline, the Methods section does not typically exceed 3,000 words. To increase reproducibility, authors are encouraged to deposit a detailed description of protocols used in their study in a protocol sharing platform of their choice. *Springer Nature's* protocols.io is a free and open service designed to help researchers share experimental know-how. Protocols deposited by the authors in www.protocols.io will be linked to the online Methods section upon publication

Detailed descriptions of methods already published should be avoided; a reference number can be provided to save space, with any new addition or variation stated.

The Methods section should be subdivided by short bold headings referring to methods used and we encourage the inclusion of specific subsections for statistics, reagents and animal models. If further references are included in this section their numbering should continue from the end of the last reference number in the rest of the paper and they are listed after the Methods section.

Please provide separate Data Availability and Code Availability statements after the main text statements and before the Extended Data legends; detailed guidance can be found in our data availability and data citations policy. Certain data types must be deposited in an appropriate public structured data depository (see detailed information), and the accession number(s) provided in the manuscript. Full access is required at the time

of publication. Should full access to data be required for peer review, authors must provide it.

The Methods section cannot contain figures or tables (essential display items should be included in the Extended Data or exceptionally in the Supplementary Information).

References

References are each numbered, ordered sequentially as they appear in the text, tables, boxes, figure legends, Methods, Extended Data tables and Extended Data figure legends.

When cited in the text, reference numbers are superscript, not in brackets unless they are likely to be confused with a superscript number.

Do not use linked fields (produced by EndNote and similar programs). Please use the one-click button provided by EndNote to remove EndNote codes before saving your file.

As a guideline, Articles allow up to 50 references in the main text if needed and within the average page budget. Only one publication can be listed for each number. Additional references for Methods or Supplementary Information are not included in this count.

Only articles that have been published or accepted by a named publication, or that have been uploaded to a recognized preprint server (for example, arXiv, bioRxiv), should be in the reference list; papers in preparation should be mentioned in the text with a list of authors (or initials if any of the authors are co-authors of the present contribution).

Published conference abstracts, numbered patents, preprints on recognized servers, papers in press, and research datasets that have been assigned a digital object identifier may be included in reference lists, but text, grant details and acknowledgements may not. (An

exception is the highlighted references which we ask authors of Reviews, Perspectives and Insights articles to provide.)

All authors should be included in reference lists unless there are more than five, in which case only the first author should be given, followed by 'et al.'

Please follow the style below in the published edition of *Nature* in preparing reference lists.

- Authors should be listed surname first, followed by a comma and initials of given names.

- Titles of all cited articles are required. Titles of articles cited in reference lists should be in upright, not italic text; the first word of the title is capitalized, the title written exactly as it appears in the work cited, ending with a full stop. Book titles are italic with all main words capitalized. Journal titles are italic and abbreviated according to common usage. Volume numbers are bold. The publisher and city of publication are required for books cited. (Refer to published papers in Nature for details.)

- Research datasets may be cited in the reference list if they have been assigned digital object identifiers (DOIs) and include authors, title, publisher (repository name), identifier (DOI expressed as a URL). Example: Hao, Z., AghaKouchak, A., Nakhjiri, N. & Farahmand, A. Global Integrated Drought Monitoring and Prediction System (GIDMaPS) data sets. figshare http://dx.doi.org/10.6084/m9.figshare.853801 (2014).

- Recognized preprints may be cited in the reference list. Example: Babichev, S. A., Ries, J. & Lvovsky, A. I. Quantum scissors: teleportation of single-mode optical states by means of a nonlocal single photon. Preprint at http://arXiv.org/quant-ph/0208066 (2002).

- References to web-only journals should give authors, article title and journal name as above, followed by URL in full - or DOI if known - and the year of publication in parentheses.

- References to websites should give authors if known, title of cited page, URL in full, and year of posting in parentheses.

End notes

End notes are brief and follow the Methods (or Methods References, if any).

Acknowledgements should be brief, and should not include thanks to anonymous referees and editors, inessential words, or effusive comments. A person can be thanked for assistance, not "excellent" assistance, or for comments, not "insightful" comments, for example.

Any relevant funding information for primary research papers should be declared in a separate funding statement. Funding should be declared only if the work in the article is within scope of the grant and arose directly from a specific grant.

Author Contributions: Authors are required to include a statement to specify the contributions of each co-author. The statement can be up to several sentences long, describing the tasks of individual authors referred to by their initials. See the authorship policy page for further explanation and examples.

Competing interests statement.

Additional Information: Authors should include a set of statements at the end of the paper, in the following order:

- Papers containing Supplementary Information contain the statement:

"Supplementary Information is available for this paper."

- A sentence reading "Correspondence and requests for materials should be addressed to XX." Nature expects this identified author to respond to readers' enquiries and requests for materials, and to coordinate the handling of any other matters arising from the published contribution, including corrections complaints. The author named as corresponding author is not necessarily the senior author, and publication of this author's name does not imply seniority. Authors may include more than one e-mail address if essential, in which event Nature will communicate with the first-listed address for any post-publication matters, and expect that author to coordinate with the other co-authors.

- Peer review information includes the names of reviewers who agree to be cited and is completed by Nature staff during proofing.

- A sentence reading "Reprints and permissions information is available at www.nature.com/reprints."

Life sciences and behavioural & social sciences reporting guidelines

To improve the transparency of reporting and the reproducibility of published results, authors of life sciences and behavioural & social sciences Articles must provide a completed Reporting Summary that will be made available to editors and reviewers during manuscript assessment. The Reporting Summary will be published with all accepted manuscripts.

Please note: because of the advanced features used in these forms, you must use Adobe Reader to open the documents and fill them out.

See guidance and resources related to the use and reporting of statistics.

Tables

Tables should each be presented on a separate page, portrait (not landscape) orientation, and upright on the page, not sideways.

Tables have a short, one-line title in bold text. Tables should be as small as possible. Bear in mind the size of a Nature page as a limiting factor when compiling a table.

Symbols and abbreviations are defined immediately below the table, followed by essential descriptive material as briefly as possible, all in double-spaced text.

Standard table formats are available for submissions of cryo-EM, NMR and X-ray crystallography data. Authors providing these data must use these standard tables and include them as Extended Data.

Figure legends

For initial submissions, we encourage authors to present the manuscript text and figures together in a single Word doc or PDF file, and for each figure legend to be presented together with its figure. However, when preparing the final paper to be accepted, we require figure legends to be listed one after the other, as part of the text document, separate from the figure files, and after the main reference list.

Each figure legend should begin with a brief title for the whole figure and continue with a short description of each panel and the symbols used. If the paper contains a Methods section, legends should not contain any details of methods. Legends should be fewer than 300 words each.

All error bars and statistics must be defined in the figure legend, as discussed above.

Figures

Nature requires figures in electronic format. Please ensure that all digital images comply with the Nature journals' policy on image integrity.

Figures should be as small and simple as is compatible with clarity. The goal is for figures to be comprehensible to readers in other or related disciplines, and to assist their understanding of the paper. Unnecessary figures and parts (panels) of figures should be avoided: data presented in small tables or histograms, for instance, can generally be stated briefly in the text instead. Avoid unnecessary complexity, colouring and excessive detail.

Figures should not contain more than one panel unless the parts are logically connected; each panel of a multipart figure should be sized so that the whole figure can be reduced by the same amount and reproduced on the printed page at the smallest size at which essential details are visible. For guidance, *Nature*'s standard figure sizes are 90 mm (single column) and 180 mm (double column) and the full depth of the page is 170 mm.

Amino-acid sequences should be printed in Courier (or other monospaced) font using the one-letter code in lines of 50 or 100 characters.

Authors describing chemical structures should use the Nature Research Chemical Structures style guide.

Some brief guidance for figure preparation:

- Lettering in figures (labelling of axes and so on) should be in lower-case type, with the first letter capitalized and no full stop.

- Units should have a single space between the number and the unit, and follow SI nomenclature or the nomenclature common to a particular field. Thousands should be separated by commas (1,000). Unusual units or abbreviations are defined in the legend.

- Scale bars should be used rather than magnification factors.

- Layering type directly over shaded or textured areas and using reversed type (white lettering on a coloured background) should be avoided where possible.

- Where possible, text, including keys to symbols, should be provided in the legend rather than on the figure itself.

Figure quality

At initial submission, figures should be at good enough quality to be assessed by referees, preferably incorporated into the manuscript text in a single Word doc or PDF, although figures can be supplied separately as JPEGs if authors are unable to include them with the text. Authors are advised to follow the initial and revised submissions guidelines with respect to sizing, resolution and labelling.

Please note that print-publication quality figures are large and it is not helpful to upload them at the submission stage. Authors will be asked for high-quality figures when they are asked to submit the final version of their article for publication.At that stage, please prepare figures according to these guidelines.

Third party rights

Nature discourages the use or adaptation of previously published display items (for example, figures, tables, images, videos or text boxes). However, we recognize that to illustrate some concepts the use of published data is required and the reuse of previously published display items may be necessary. Please note that in these instances we might not be able to obtain the necessary rights for some images to be reused (as is, or adapted versions) in our articles. In such cases, we will contact you to discuss the sourcing of alternative material.

Figure costs

In order to help cover some of the additional cost of four-colour reproduction, Nature Portfolio charges our authors a fee for the printing of their colour figures. Please contact our offices for exact pricing and details. Inability to pay this charge will not prevent publication of colour figures judged essential by the editors, but this must be agreed with the editor prior to acceptance.

Production-quality figures

When a manuscript is accepted in principle for publication, the editor will ask for high-resolution figures. Do not submit publication-quality figures until asked to do so by an editor. At that stage, please prepare figures according to these guidelines.

Extended Data

Extended Data figures and tables are online-only (appearing in the online PDF and full-text HTML version of the paper), peer-reviewed display items that provide essential background to the Article but are not included in the printed version of the paper due to space constraints or being of interest only to a few specialists. A maximum of ten Extended Data display items (figures and tables) is typically permitted. See Composition of a Nature research paper.

Extended Data tables should be formatted along similar lines to tables appearing in print (see section 5.7) but the main body (excluding title and legend, which should be included at the end of the Word file) should be submitted separately as an image rather than as an editable format in Word, as Extended Data tables are not edited by Nature's subediting department. Small tables may also be included as sub-panels within Extended Data figures. See Extended Data Formatting Guide.

Extended Data figures should be prepared along slightly different guidelines compared to figures appearing in print, and may be multi-panelled as long as they fit to size rules (see Extended Data Formatting Guide). Extended Data figures are not edited or styled by Nature's art

department; for this reason, authors are requested to follow Nature style as closely as possible when preparing these figures. The legends for Extended Data figures should be prepared as for print figures and should be listed one after the other at the end of the Word file.

If space allows, Nature encourages authors to include a simple schematic, as a panel in an Extended Data figure, that summarizes the main finding of the paper, where appropriate (for example, to assist understanding of complex detail in cell, structural and molecular biology disciplines).

If a manuscript has Extended Data figures or tables, authors are asked to refer to discrete items at an appropriate place in the main text (for example, Extended Data Fig. 1 and Extended Data Table 1).

If further references are included in the Extended Data tables and Extended Data figure legends, the numbering should continue from the end of the last reference number in the main paper (or from the last reference number in the additional Methods section if present) and the list should be added to the end of the list accompanying the additional Methods section, if present, or added below the Extended Data legends if no additional Methods section is present.

Supplementary Information

Supplementary Information (SI) is online-only, peer-reviewed material that is essential background to the Article (for example, large data sets, methods, calculations), but which is too large or impractical, or of interest only to a few specialists, to justify inclusion in the printed version of the paper. See the Supplementary Information page for further details.

Supplementary Information should not contain figures (any figures additional to those appearing in print should be formatted as Extended Data figures). Tables may be included in Supplementary Information, but only if they are unsuitable for formatting as Extended Data tables (for

example, tables containing large data sets or raw data that are best suited to Excel files).

If a manuscript has accompanying SI, either at submission or in response to an editor's letter that requests it, authors are asked to refer to discrete items of the SI (for example, videos, tables) at an appropriate point in the main manuscript.

Chemical structures and characterization of chemical materials

For guidelines describing *Nature*'s standards for experimental methods and the characterization of new compounds, please see the information sheet on the characterization of chemical materials.

We aim to produce chemical structures in a consistent format throughout our articles. Please use the Nature Portfolio Chemical Structures Guide and ChemDraw template to ensure that you prepare your figures in a format that will require minimal changes by our art and production teams. Submit final files at 100% as .cdx files.

Registered Reports

Registered Reports are empirical articles testing confirmatory hypotheses in which the methods and proposed analyses are pre-registered and peer reviewed prior to research being conducted. For further details about Registered Reports and instructions for how to submit such articles to *Nature* please consult our Registered Reports page.

Submission

All contributions should be submitted online, unless otherwise instructed by the editors. Please be sure to read the information on what to include in your cover letter as well as several important content-related issues when putting a submission together.

Before submitting, all contributors must agree to all of Nature's publication policies.

Nature authors must make data and materials publicly available upon publication. This includes deposition of data into the relevant databases and arranging for them to be publicly released by the online publication date (not after). View a description of our initiative to improve the transparency and the reproducibility of published results. A full description of Nature's publication policies is at the Nature Portfolio Authors and Referees website.

Other Nature Research journals

An account of the relationship between all the Nature journals is provided at the Nature family page.

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